

Powers of Ten

Every graded set, every week — 4 weeks, 20 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Ten Times Bigger

Every place is ten of the place to its right. Say it as you write it.

Warm-Up 6 ITEMS

The 4 in 40 is worth _____

The 4 in 400 is worth _____

Ten times 30 _____

Ten times 700 _____

One tenth of 500 _____

One tenth of 60 _____

Core GRADED • SHOW YOUR WORKING

1. The 7 in 7,000 is how many times the 7 in 700 _____

2. The 3 in 300 is how many times the 3 in 3 (Two jumps left) _____

3. 100 times 40 _____

4. One hundredth of 900 _____

5. The 5 in 50,000 is worth _____

★ Challenge NOT GRADED

C1. The 6 in 60,000 is how many times the 6 in 60 (Three jumps) _____

C2. A digit moves two places left. Its value is multiplied by _____

Down to Thousandths

Three places past the point. Type decimals with a leading zero, like 0.45.

Warm-Up 6 ITEMS

Write four tenths as a decimal

Write seven hundredths

Write three thousandths

Write twenty-five hundredths

The 5 in 0.5 means five

Write one half as a decimal

Core GRADED • SHOW YOUR WORKING

1. Write four hundred six thousandths

2. Write two and thirty-five hundredths

3. 0.60 written more simply

4. Write sixty-two thousandths

5. How many hundredths in 0.3

Challenge NOT GRADED

C1. Write nine and nine thousandths

C2. How many thousandths in 0.25

Powers of Ten

The digits move. The point stays put.

Warm-Up 6 ITEMS

3.5×10

3.5×100

$42 \div 10$

$42 \div 100$

10^2 as a number

10^3 as a number

Core GRADED • SHOW YOUR WORKING

1. 0.06×100 _____

2. $7 \div 1000$ _____

3. 2.4×10^3 _____

4. $850 \div 10^2$ _____

5. 10^4 as a number _____

Challenge NOT GRADED

C1. 0.035×10^4

C2. Which power of ten takes 0.7 to 700 (Three places)

Compare & Round

Line up the points. Trailing zeros cost nothing.

✦ Warm-Up 6 ITEMS

Larger: 0.5 or 0.35 — type it

Larger: 0.7 or 0.70

Round 4.7 to the nearest whole

Round 0.34 to the nearest tenth

Round 128 to the nearest ten

Round 0.86 to the nearest tenth

◆ Core GRADED • SHOW YOUR WORKING

1. Round 2.451 to the nearest hundredth

.....

2. Round 9.96 to the nearest tenth

.....

3. Larger: 0.406 or 0.41

.....

4. Round 0.0475 to the nearest thousandth

.....

5. Round 4,829 to the nearest hundred

.....

★ Challenge NOT GRADED

C1. A number rounds to 5.3 to the nearest tenth. Type its smallest possible value

.....

C2. Round 0.999 to the nearest hundredth

.....

Decimal Duel & Zoom Atlas

Enrichment. Build big, then justify it.

★ Warm-Up 2 ITEMS

Biggest number from digits 4, 7, 1, 9

Smallest number from those digits

◆ Core GRADED • SHOW YOUR WORKING

1. Biggest decimal under 1 from 4, 7, 1 — type it

.....

2. A model is 100 times a 3 cm object. Its size in cm

.....

★ Challenge NOT GRADED

C1. An object is drawn one thousandth of true size. True length 4000 mm, so drawn length in mm

C2. Digits 2, 5, 8: the value closest to 1 you can build as 0.____ (Biggest is nearest to one)

C3. How many times bigger is 8.5 than 0.085

Week 1 Answers

Mission 01 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • Ten Times Bigger

OUT OF 11

Warm-Up • 40 | 400 | 300 | 7000 | 50 | 6

Core • 10 | 100 | 4000 | 9 | 50000

Challenge • 1000 | 100

1.2 • Down to Thousandths

OUT OF 11

Warm-Up • 0.4 | 0.07 | 0.003 | 0.25 | tenths | 0.5

Core • 0.406 | 2.35 | 0.6 | 0.062 | 30

Challenge • 9.009 | 250

1.3 • Powers of Ten

OUT OF 11

Warm-Up • 35 | 350 | 4.2 | 0.42 | 100 | 1000

Core • 6 | 0.007 | 2400 | 8.5 | 10000

Challenge • 350 | 1000

1.4 • Compare & Round

OUT OF 11

Warm-Up • 0.5 | equal | 5 | 0.3 | 130 | 0.9

Core • 2.45 | 10 | 0.41 | 0.048 | 4800

Challenge • 5.25 | 1

Fri • Decimal Duel & Zoom Atlas

OUT OF 4

Warm-Up • 9741 | 1479

Core • 0.741 | 300

Challenge • 4 | 0.852 | 100

Multiply by Ten

The digits move. The point stays exactly where it was.

✦ Warm-Up 6 ITEMS

4.2×10

4.2×100

0.7×10

0.7×100

3×10

3×1000

◆ Core GRADED • SHOW YOUR WORKING

1. 0.06×100 _____

2. 2.4×1000 _____

3. 0.035×100 _____

4. 12.5×10 _____

5. 0.008×1000 _____

★ Challenge NOT GRADED

C1. 0.0405×1000

C2. Which power of ten takes 0.7 to 7000

Divide by Ten

The same journey in reverse.

Warm-Up 6 ITEMS

$42 \div 10$

$420 \div 100$

$7 \div 10$

$70 \div 100$

$300 \div 10$

$5000 \div 1000$

Core GRADED • SHOW YOUR WORKING

1. $6 \div 100$

.....

2. $850 \div 100$

.....

3. $7 \div 1000$

.....

4. $125 \div 10$

.....

5. $9 \div 10$

.....

Challenge NOT GRADED

C1. $40.5 \div 1000$

.....

C2. $0.5 \div 100$

.....

Exponent Shorthand

10^3 means three tens multiplied, which is three places.

Warm-Up 6 ITEMS

 10^1 10^2 10^3 10^4 Zeros in 10^5 10^6

Core GRADED • SHOW YOUR WORKING

1. 3×10^2

2. 4.5×10^3

3. 7×10^4

4. 0.6×10^3

5. 2.5×10^2

Challenge NOT GRADED

C1. $10^3 \times 10^2$

C2. 0.035×10^4

Patterns in Zeros

Why 4×100 ends in two zeros and 0.4×100 does not.

Warm-Up 6 ITEMS

4×100

0.4×100

40×100

0.04×100

Zeros at the end of 4×1000

0.4×1000

Core GRADED • SHOW YOUR WORKING

1. 25×100

2. 0.25×100

3. 0.025×100

4. 1.5×1000

5. 0.015×1000

Challenge NOT GRADED

C1. 0.0025×10000

.....

C2. How many times bigger is 8.5 than 0.085

.....

Zoom Atlas Begins

Pick the object. Draw it at true size first.

★ Warm-Up 2 ITEMS

A 3 cm object at 10 times — cm

At 100 times — cm

◆ Core GRADED • SHOW YOUR WORKING

1. A 4 m object at one hundredth — cm

2. A 250 cm object at one tenth — cm

★ Challenge NOT GRADED

C1. A 4000 mm object at one thousandth — mm

C2. A 2.5 cm object at 1000 times — metres

C3. From 0.5 mm to 5 m — the multiplier

Week 2 Answers

Mission 01 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • Multiply by Ten

OUT OF 11

Warm-Up • 42 | 420 | 7 | 70 | 30 | 3000

Core • 6 | 2400 | 3.5 | 125 | 8

Challenge • 40.5 | 10000

2.2 • Divide by Ten

OUT OF 11

Warm-Up • 4.2 | 4.2 | 0.7 | 0.7 | 30 | 5

Core • 0.06 | 8.5 | 0.007 | 12.5 | 0.9

Challenge • 0.0405 | 0.005

2.3 • Exponent Shorthand

OUT OF 11

Warm-Up • 10 | 100 | 1000 | 10000 | 5 | 1000000

Core • 300 | 4500 | 70000 | 600 | 250

Challenge • 100000 | 350

2.4 • Patterns in Zeros

OUT OF 11

Warm-Up • 400 | 40 | 4000 | 4 | 3 | 400

Core • 2500 | 25 | 2.5 | 1500 | 15

Challenge • 25 | 100

Fri • Zoom Atlas Begins

OUT OF 4

Warm-Up • 30 | 300

Core • 4 | 25

Challenge • 4 | 25 | 10000

Round to a Named Place

Find the place, look one to its right, decide.

✦ Warm-Up 6 ITEMS

Round 47 to the nearest ten

Round 43 to the nearest ten

Round 250 to the nearest hundred

Round 4.7 to the nearest whole

Round 3.2 to the nearest whole

Round 9.8 to the nearest whole

◆ Core GRADED • SHOW YOUR WORKING

1. Round 4,829 to the nearest thousand

2. Round 4,829 to the nearest hundred

3. Round 12.5 to the nearest whole

4. Round 0.48 to the nearest tenth

5. Round 2.451 to the nearest hundredth

★ Challenge NOT GRADED

C1. Round 9.96 to the nearest tenth

C2. Round 0.0475 to the nearest thousandth

Rounding Decimals

Same method, smaller places.

✦ Warm-Up 6 ITEMS

Round 0.4 to the nearest whole

Round 0.6 to the nearest whole

Round 1.5 to the nearest whole

Round 2.4 to the nearest whole

Round 0.35 to the nearest tenth

Round 0.24 to the nearest tenth

◆ Core GRADED • SHOW YOUR WORKING

1. Round 3.456 to the nearest hundredth

2. Round 3.456 to the nearest tenth

3. Round 0.075 to the nearest hundredth

4. Round 12.349 to the nearest tenth

5. Round 7.005 to the nearest hundredth

★ Challenge NOT GRADED

C1. Round 0.9999 to the nearest thousandth

C2. Round 2.4999 to the nearest whole

Which Place Matters

Money rounds to hundredths. Distance rarely does.

 Warm-Up 6 ITEMS

Round \$4.56 to the nearest cent

Round \$4.567 to the nearest cent

Round 4.7 to the nearest whole

Round 250 to the nearest hundred

Round 0.5 to the nearest whole

Round \$9.49 to the nearest dollar

 Core GRADED • SHOW YOUR WORKING

1. Round \$19.99 to the nearest dollar

2. Round 1.005 m to the nearest centimetre

3. Round 47 minutes to the nearest ten

4. 4.4 buses of people — buses needed

5. Round 2.44 km to the nearest tenth

 Challenge NOT GRADED

C1. Round \$0.075 to the nearest cent

C2. A price of \$2.994 charged to the cent

Estimate to Check

Round first, compute second, compare.

 Warm-Up 6 ITEMS

Estimate $4.9 + 3.1$

Estimate $9.8 - 4.9$

Estimate 2.1×4

Estimate $11.9 \div 4$

Estimate $0.9 + 1.1$

Estimate 5.2×2

 Core GRADED • SHOW YOUR WORKING

1. Estimate 412×19

.....
2. Estimate 0.49×100

.....
3. True value of 0.49×100

.....
4. Estimate $19.8 \div 5$

.....
5. Estimate $12.4 - 7.9$

.....

 Challenge NOT GRADED

C1. Estimate $5,120 \times 48$

.....

C2. True value of $12.4 - 7.856$

.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

 Warm-Up 2 ITEMS

The 7 in 7,000 is how many times the 7 in 700

Write four hundred six thousandths

 Core GRADED • SHOW YOUR WORKING

1. 0.06×100
2. $850 \div 10^2$
3. Write 10^4 as an ordinary number
4. Round 2.451 to the nearest hundredth
5. Larger: 0.406 or 0.41

 Challenge NOT GRADED

C1. Which power of ten takes 0.7 to 7,000

.....

Week 3 Answers

Mission 01 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Round to a Named Place

OUT OF 11

Warm-Up • 50 | 40 | 300 | 5 | 3 | 10

Core • 5000 | 4800 | 13 | 0.5 | 2.45

Challenge • 10 | 0.048

3.2 • Rounding Decimals

OUT OF 11

Warm-Up • 0 | 1 | 2 | 2 | 0.4 | 0.2

Core • 3.46 | 3.5 | 0.08 | 12.3 | 7.01

Challenge • 1 | 2

3.3 • Which Place Matters

OUT OF 11

Warm-Up • 4.56 | 4.57 | 5 | 300 | 1 | 9

Core • 20 | 1.01 | 50 | 5 | 2.4

Challenge • 0.08 | 2.99

3.4 • Estimate to Check

OUT OF 11

Warm-Up • 8 | 5 | 8 | 3 | 2 | 10

Core • 8000 | 50 | 49 | 4 | 4

Challenge • 250000 | 4.544

Fri • Mid-Unit Quiz

OUT OF 7

Warm-Up • 10 | 0.406

Core • 6 | 8.5 | 10000 | 2.45 | 0.41

Challenge • 10000

Finish the Atlas

Every jump labelled with its power of ten.

Warm-Up 6 ITEMS

A 3 cm object at 10 times — cm

At 100 times — cm

A 4 m object at one hundredth — cm

A 250 cm object at one tenth — cm

0.5 mm at 10 times — mm

A 2.5 cm object at 100 times — cm

Core GRADED • SHOW YOUR WORKING

1. A 4,000 mm object at one thousandth — mm

2. A 2.5 cm object at 1000 times — metres

3. From 0.5 mm to 5 m — the multiplier

4. 3.5 m at 100 times — metres

5. 0.035×10^4

Challenge NOT GRADED

C1. 0.0405×1000

C2. A model one thousandth of 4,000 mm — mm

Explain a Jump

Pick one panel and say what moved and what stayed.

Warm-Up 6 ITEMS

4.2×10

$42 \div 10$

0.7×100

$70 \div 100$

3×1000

$3000 \div 1000$

Core GRADED • SHOW YOUR WORKING

1. 0.06×100 _____

2. $6 \div 100$ _____

3. 2.4×1000 _____

4. $2400 \div 1000$ _____

5. 0.008×1000 _____

Challenge NOT GRADED

C1. $40.5 \div 1000$

C2. How many times bigger is 8.5 than 0.085

Mixed Review

Place value, powers and rounding in one set.

Warm-Up 6 ITEMS

The 4 in 400 is how many times the 4 in 40

One tenth of 500

10^3

Zeros in 10^5

Larger: 0.5 or 0.35

Round 4.7 to the nearest whole

Core GRADED • SHOW YOUR WORKING

1. 3.5×100

2. $42 \div 100$

3. 0.035×100

4. Round 4,829 to the nearest thousand

5. Smallest of 0.6, 0.55, 0.506, 0.65

Challenge NOT GRADED

C1. 0.0025×10000

C2. Round 2.451 to the nearest hundredth

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 6 ITEMS

Tenths in 0.3

10^2

4.2×10

$7 \div 1000$

Is 0.4 the same as 0.40 — yes or no

0.05 — name the last place

Core GRADED • SHOW YOUR WORKING

1. $0.125 + 0.875$

2. $850 \div 100$

3. 0.06×100

4. Round 9.96 to the nearest tenth

5. Larger: 0.08 or 0.075

Challenge NOT GRADED

C1. $12.4 - 7.856$

C2. 0.0405×1000

Mission 01 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 2 ITEMS

The 3 in 300 is how many times the 3 in 3

Write sixty-two thousandths

Core GRADED • SHOW YOUR WORKING

1. Hundredths in 0.3

.....

2. 2.4×10^3

.....

3. $7 \div 1000$

.....

4. Round 9.96 to the nearest tenth

.....

5. Round 0.0475 to the nearest thousandth

.....

6. Which power of ten takes 0.7 to 700

.....

7. Smallest of 0.6, 0.55, 0.506, 0.65

.....

8. 0.035×10^4

.....

Challenge NOT GRADED

C1. $3 + 0.4 + 0.007$ as one decimal

C2. A model one thousandth of 4,000 mm — mm

Week 4 Answers

Mission 01 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 • Finish the Atlas

OUT OF 11

Warm-Up • 30 | 300 | 4 | 25 | 5 | 250

Core • 4 | 25 | 10000 | 350 | 350

Challenge • 40.5 | 4

4.2 • Explain a Jump

OUT OF 11

Warm-Up • 42 | 4.2 | 70 | 0.7 | 3000 | 3

Core • 6 | 0.06 | 2400 | 2.4 | 8

Challenge • 0.0405 | 100

4.3 • Mixed Review

OUT OF 11

Warm-Up • 10 | 50 | 1000 | 5 | 0.5 | 5

Core • 350 | 0.42 | 3.5 | 5000 | 0.506

Challenge • 25 | 2.45

Thu • Error Journal Sweep

OUT OF 11

Warm-Up • 3 | 100 | 42 | 0.007 | yes | hundredths

Core • 1 | 8.5 | 6 | 10 | 0.08

Challenge • 4.544 | 40.5

Fri • Mission 01 Test

OUT OF 10

Warm-Up • 100 | 0.062

Core • 30 | 2400 | 0.007 | 10 | 0.048 | 1000 | 0.506 | 350

Challenge • 3.407 | 4